Legalyze: An Open-Source Structure-Aware Agentic RAG Platform for High-Precision Legal Intelligence

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ABSTRACT

Artificial Intelligence holds significant potential for transforming legal workflows, yet the adoption of Large Language Models (LLMs) in statutory law remains constrained by domain-specific hallucinations and lack of verifiable citations. In this paper, we introduce Legalyze, an open-source, structure-aware agentic Retrieval-Augmented Generation (RAG) platform tailored for Pakistani jurisprudence. Legalyze features Structure-Aware Hierarchical Indexing (SAHI) to preserve statutory hierarchies across constitutional, criminal, and procedural law, alongside a dual-stream hybrid retriever (BM25 + Dense Semantic Embeddings) integrated via Reciprocal Rank Fusion (RRF). A multi-agent reasoning architecture coordinates structured statutory retrieval, cross-statute verification, and citation grounding.

We evaluate Legalyze using a reviewer-blinded protocol with three independent High Court Advocates across 67 legal queries spanning 1.2 million tokens. Legalyze achieves a 5.5% pooled hallucination rate (4.5% consensus-adjudicated), representing a 90.0% relative reduction over Zero-Shot GPT-4o (54.7%) and a 76.0% reduction over Standard Flat RAG (22.9%). All pairwise comparisons are confirmed statistically significant by paired inferential testing (Cochran's $Q = 37.57$, $p = 6.93 \times 10^{-9}$; GEE odds ratio: 20.88× over Zero-Shot). Legalyze offers a reproducible blueprint for domain-adapted, verifiable legal intelligence.

Key Methodological Novelties:

- SAHI Parser: Hierarchical regex-based statutory boundary indexing.
- Dual-Stream Hybrid RAG: BM25 + Dense Vectors combined via RRF.
- Multi-Agent Orchestration: Retrieval, reasoning & verification agents.

Empirical & Statistical Rigor:

- 1.2 Million Tokens: 5 primary Pakistani statutes & court precedents.
- 603 Blinded Judgments: 3 High Court Advocates across 67 queries.
- Significant Reduction: Cochran's $Q = 37.57$ ($p = 6.93 \times 10^{-9}$).